A Hybrid Neural Network and Modified Pro-Energy Approach for Solar Energy Prediction in Wireless Sensor Networks

***Abstract*—The wireless sensor network (WSN) has a variety of applications in the real world. The greatest challenge in these networks is powering the sensor nodes for an extended period without needing any manual assistance. One popular method to overcome this limitation is harvesting solar energy; however, solar energy is dependent on three primary factors: weather, time of day, and the season. Therefore, accurate forecasting of incoming energy from these multiple factors is critical for effective power management in a WSN.**

This study evaluates and compares five different energy forecast methods: an exponentially weighted moving average (EWMA), a weighted cumulative moving average (WCMA), Pro-Energy, a modified version of Pro-Energy (MPE), and a CNN-LSTM (convolutional neural network-long short term memory) method with actor-critic reinforcement learning (CNN-LSTM-AC). We also propose a hybrid model that combines MPE with a lightweight multi-layer perceptron (MLP) neural network. The input data to the MLP includes yesterday's solar energy, and the rolling median and maximum from the past seven days, plus a time-shifted energy term and cyclic time-of-day characteristics; thus, the MLP learns patterns that rule-based models cannot identify. The final prediction estimate is comprised of 80% MPE and 20% MLP.

We assessed all models using the same NREL solar irradiance data set; we used the Prediction Error Ratio (PER), which is identical to the one that Cammarano and colleagues used when presenting their Pro-Energy algorithm [15]. The average PER for the hybrid model across all five test days was 0.7822, which gave it the best PER value. Although the CNN-LSTM-AC model is much more complicated than the hybrid model, its average PER value across the five testing days was 97.9634 because it was not able to accurately predict energy production on Days 100 and 300, yielding PER values of 365.61 and 121.99 for those days, respectively. On those same two days, the hybrid model yielded PER values of 0.9629 and 0.9054. This suggests that self-correction within a single day by MPE is far more important in providing consistent performance than is just the model's complexity level.

***Index Terms*—Wireless Sensor Network, Solar Energy Harvesting, Energy prediction, Modified Pro-Energy, Artificial Neural Network, CNN-LSTM, Actor-Critic, Hybrid model, Prediction Error Ratio.**

I. Introduction

Wireless sensor networks (WSNs) are made up of numerous powered sensor nodes that are distributed throughout an area. These nodes can gather data from many different types of sources with different types of sensors, including temperature, light, humidity, and pressure. Applications for WSNs include agriculture, border patrols, smart cities, industrial controllers, and more [1], [2]. A WSN sensor node has four major components: a transceiver, sensing, processing, and power unit. The power component is the most important, since if it runs out of battery power, the entire sensor node becomes inoperable [3]. Since most traditional types of WSN sensor nodes use disposable batteries, it is necessary to manually replace the battery once it runs out of power, which creates a significant burden for those nodes located in remote, inaccessible locations [4].

One way to solve this problem is through energy harvesting, which collects energy from the environment and uses that harvested energy to recharge each node's battery so that it can operate indefinitely. Photovoltaic solar energy harvesting is one of the most researched strategies for power in WSN applications. Since photovoltaic panels are low-cost, lightweight, and easily accessible, solar power has quickly become one of the most preferred methods for powering WSN applications over the long term [5].

Solar irradiance is not constant and varies throughout the day, across seasons, and depending on weather conditions such as cloud cover and rain. Thus, the difficulty arises for the node in planning energy consumption consistently. A node will deplete its battery if it consumes more than it can harvest, while it is wasting potential resources if it consumes less than it can harvest [6]. The effect of a prediction error is not balanced; generally, the penalty associated with over-predicting the amount of energy will be higher than that associated with under-predicting the amount of energy for battery-powered nodes since the former will degrade the battery while the latter will simply mean that the battery stores unutilised energy. An ongoing pattern of excessive overestimating will result in the battery reaching the minimum threshold and causing the node to go into a shutdown state, which negates the potential benefits of energy harvesting entirely.

Energy prediction is accomplished by predicting the amount of solar energy that a node will receive during the next time period. Nodes that can accurately predict receive timely scheduling for transmissions when there is enough energy available, extend sleep periods when there is little energy available, and decrease shutdowns due to loss of power [7], [8]. Predicting with greater accuracy helps to increase network lifetime and make systems more reliable. There are three different time scales for the variation of solar irradiance due to: the sun's position in the sky, the weather patterns of different days, and the position of the sun changing throughout the seasons. Any prediction method needs to be able to account for all three of these variations simultaneously with historical observations as input only.

Profile-based methods such as Pro-Energy store daily shapes of irradiance and find the best daily historical match to each new day. This works well if the current day looks like a historic day, but it does not work at all if the current day is different from those in history. Exponential smoothing methods such as EWMA are appropriate for short-term change but do not work very well for longer-term change. Deep learning approaches can, in theory, learn all three of these scales at once but require a lot of training data, computational resources, and do not have a built-in method for recovering from errors that accumulate within the same day. The hybrid model proposed in this paper is based on the observation that MPE's within-day correction is the most successful approach for maintaining a low PER across a variety of conditions, and a lightweight learning component can improve it on days when the historical profile matching is weak.

In addition to describing new methods of predicting future solar irradiance, this paper implements five separate techniques: EWMA, WCMA, Pro-Energy, Modified Pro-Energy (MPE), and CNN-LSTM-AC as prediction algorithms, and discusses a hybrid approach that combines the MPE model with a small MLP. The dataset and metric used in this study are the same as those used by Cammarano et al. [15] for the original Pro-Energy work and by Amgoth and Jana [10] for their energy management solutions in wireless sensor networks, allowing direct comparison across all methods.

The structure of this paper follows: Section II describes related work, Section III describes the problem, Section IV describes the methods, Section V describes the dataset, Section VI describes the experimental setup, Section VII describes the results, Section VIII discusses the findings, and Section IX concludes this work.

II. Related Work

A. Statistical and Profile-Based Methods

Kansal et al. [9] established an energy-neutral operation framework that proved a node is able to function indefinitely as long as the energy consumption is always within the extent of the energy generated. Thus, this work provided the theoretical basis for energy-neutral system operation and demonstrated the importance of accurate energy prediction. A good prediction will enable the node to schedule all of its workload precisely to the maximum generated, while if the prediction is inaccurate, the node needs to have a larger safety factor and will therefore be wasting capacity. This framework has been widely cited in the literature as the foundation for energy management in wireless sensor networks.

ASIM was developed by Ashraf et al. [12] as a method for predicting energy use by using data that have daily variation; it is different from using an hourly variance such as those collected with Pro-Energy. ASIM's main strength is that it provides uncertainty in its predictions; however, it does not produce a precise prediction for each time slot. Yang et al. [13] used a Weather-Conditioned Exponentially Weighted Moving Average (WC-EWMA) model to predict solar power energy use within hostile environments by modelling both battery and load profiles offline, and demonstrated that conditioning the weight on weather state information would reduce prediction lags associated with abrupt changes in solar radiation, which is similar in concept to how the GAP factor within WCMA operates.

The Pro-Energy approach, presented by Cammarano et al. [15], utilises daily profiles of energy produced to calculate short- and medium-term predictions by modelling historical data. A significant limitation of this technique is the omission of potential miss-forecasting errors accumulated throughout the duration of a given day. As demonstrated in our results, this deficiency can potentially lead to disastrous outcomes; on Day 100, Pro-Energy generated a PER of 1189.86 solely due to day-of-irradiance not corresponding with any of the existing profiles, with no opportunity for in-day correction available. The Modified Pro-Energy component in our hybrid solution is designed to remedy this deficit.

Kosunalp [16] employs a Q-learning algorithm in order to develop an adaptive secondary energy prediction system (QL-SEP) that significantly outperforms traditional moving average models throughout all seasons. An essential feature of the Q-learning agent is that it continuously updates its policy based on new observations, rather than forcing the user to specify a fixed-length smoothing window. Q-learning is a type of reinforcement learning, so it does not provide an estimate of irradiance at any given slot. Peng and Low [19] proposed a budgeting-only energy-neutral framework with no predictive elements, establishing a lower bound on what can be achieved without predictions. Gorlatova et al. [20] developed stochastic forecasts of solar energy based on historical radiation, finding that temporal patterns in solar energy data can help establish future solar energy availability. Their work on the autocorrelation of multi-day solar energy data is particularly important for Pro-Energy and MPE in choosing window lengths for profiles; a 10-day profile window captures the dominant autocorrelation scale for the majority of weather conditions.

Dondi et al. [24] conducted research to model and optimise a solar energy harvester for self-powered WSNs and then empirically characterised how variables such as panel tilt, shading, and battery capacity contributed to predictive accuracy. This research led to the conclusion that when performing a forecast of available solar energy for a WSN, broadband irradiance should be used instead of another type of directional measurement; this study has therefore selected the Global CMP22 channel for its purpose. Dehwah et al. [17] proposed an alternative method of energy management in solar-powered sensor networks through a distributed routing scheme, indicating that using a prediction-based routing scheme can extend a network's operational lifetime compared to using a routing scheme that does not consider future energy availability. Ganjewar et al. [18] proposed the hierarchical fractional LMS prediction method for reducing WSN traffic at each of its sensors. Their adaptive filter structure is similar to the EWMA model presented in this study, with the key difference being that the fractional update rule captures long-term dependencies in an irradiance signal.

B. Reinforcement Learning Techniques

Hsu et al. [14] studied the application of reinforcement learning to control duty cycles in sensor networks utilising energy harvesting. The reinforcement learner is able to classify the level of energy being harvested into three classifications: low, medium, or high. The results of the simulation demonstrated improved battery reserves and extended network lifetime without relying on an explicit forecast of the amount of energy being harvested. Kosunalp [23] expanded upon the Q-learning approach in regards to using wind energy in wireless sensor networks and demonstrated that through adaptive parameters multiple types of renewable energy sources can be supported. Tsiropoulou et al. [25] studied coalition formation and resource allocation in smart IoT applications with energy constraints, and showed that using multi-agent systems can effectively coordinate the energy usage among disparate nodes in a sensor network, but they still require a forecast of the amount of energy harvested by a node to be effective.

The primary disadvantage of using reinforcement learning for this problem area is that reinforcement learning cannot produce a numerical forecast of the energy that will be received. Therefore, in order to schedule energy-aware transmissions or sleep periods accurately, a specific number must be provided. Reinforcement Learning alone cannot produce this information, but the Actor-Critic component within the CNN-LSTM-AC does provide it because it is a post-processing refinement to the numerical forecast rather than using it as the primary source of decision-making.

C. Neural Networks and Deep Learning

Because of the time dependency of the solar energy forecasting process, recurrent neural networks, especially Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), were chosen as forecasting methods for this application [27], [28]. For example, Yu et al. [27] found with their testing of LSTMs on weather data that LSTM-based forecasting techniques perform better on complex weather data compared to statistical methods such as regression. Huang et al. [28] evaluated multiple configurations of LSTMs for hour-ahead solar energy forecasting and determined that input window length had a more significant impact on forecasting performance than the depth of the LSTM model, which is in agreement with our finding that a 15-day rolling training window had more impact than the specific size of the hidden layers. Zou et al. [22] used machine learning to design intelligent controllers for solar-powered WSNs and demonstrated that the learned predictions of the model could be used to make better duty-cycle decisions.

Hybrid models based on CNN and recurrent architectures have been found to outperform standalone recurrent models [29], [30]. Alharkan et al. [29] used a dual-stream CNN-LSTM architecture to show that simultaneous extraction of local and global temporal features provides an improvement in forecasting on days with mixed cloud cover. Sabri and El Hassouni [30] utilised a GRU-CNN combination that was competitive with the performance of short-term photovoltaic forecasts. Nguyen et al. [26] constructed an ensemble of CNN and LSTM architectures with an Actor-Critic refining layer and reported a MAPE of 13.78% in a one-hour-ahead predictive study using the same NREL dataset as this study. While the previous study used MAPE, which is a relative error metric based on averaging proportional deviations across all time slots, the metric used in this work and in the original Pro-Energy paper [15] (PER) uses the same type of relative deviation measured over the complete day including low-irradiance periods near sunrise and sunset; however, PER is calculated on a per time-slot basis and then averaged over the entire day. PER is adversely affected by time slots where the irradiance is near zero; therefore, a model that performs well under MAPE in a clean grid-connected environment may generate very high PER values when applied to a minute-level WSN dataset that contains near-zero irradiance time periods.

Deep learning models like CNN-LSTM have several real-world limitations when deployed to WSNs. Sensor nodes have limited processing and memory constraints, making it impossible to train deep learning networks on the WSN node itself. Deep learning sequence models require a long history to converge, and are more likely to fail when the target day's pattern does not closely resemble previous training patterns.

Herrería-Alonso et al. [21] found that energy prediction can be improved when using cyclic time-of-day features based on the sine and cosine of the slot index, reducing the need for geographical metadata such as latitude and elevation. Our method uses the same feature encoding. Li et al. [11] found that weather forecast data can further improve energy harvesting sensor systems in terms of task scheduling when available. The current work does not use external weather forecasts as part of the predictor, because weather forecasts may not be available at all deployment sites; making a self-contained historical irradiance measurement-based predictor is therefore the goal of this work.

This paper presents a hybrid approach for predicting both typical days and difficult edge cases. MPE makes most of the day's predictions and corrects errors within the same day through profile-based matching and error correction methods. The MLP also provides predictive information using slot-level features, which helps on days for which no suitable historical data is available. The total number of model parameters is kept below 600 and no GPU is required, making it deployable in a gateway-assisted WSN architecture where the gateway retrains the MLP daily and broadcasts updated weights to the sensor nodes.

III. Problem Statement

A. Problem Formulation

The objective is to forecast the solar energy $\hat{E}(d, n)$ that a sensor node will register at time slot n on day d, incorporating all energy data from previous time slots on that day and all data from previous days. Making an inaccurate forecast leads to direct consequences: overestimating energy means a node utilises more than it is capable of earning and depletes the battery, while underestimating means the node sits idle when it could be transmitting useful data. Because many nodes exist in remote locations without manual intervention, the system has to independently adjust to weather changes and seasonal changes.

Any prediction errors will continue to affect battery state, so errors accumulate over time. For example, if the node overestimates energy supplied by a small amount at each time slot, while the individual slot may appear to be insignificant, when there are 1440 time slots in one day, this can accumulate to a considerable energy deficit. PER captures this by measuring the relative deviation at every slot and averaging it across the day. A lower PER means less accumulated error and more reliable operation. Formally, a relative error of 1.0 indicates the average prediction at all time slots was incorrectly predicted by 100%; a PER of 0.78 (the hybrid's result) means an average relative deviation of 78%, concentrated primarily in the near-zero boundary slots at sunrise and sunset.

B. Solar Harvesting Model

Each sensor node contains a solar cell connected to a battery via a power management unit. The amount of energy captured by the solar cell is correlated with the intensity of sunlight, the angle of the sun, and the location of the node. A day is divided into N equal time slots. The energy harvested at slot n on day d is $H(d, n)$. The predicted energy for that slot is $\hat{E}(d, n)$. The following assumptions are made: sensors will be located outdoors, each sensor has the same initial energy level E_0, and predictions span short to medium time horizons from a few minutes up to several hours.

Solar irradiance exhibits a characteristic daily profile, which is useful for prediction. On an ideal clear day, the profile of solar irradiance rises to a maximum near solar noon, then returns to zero and generally conforms to a symmetric distribution about solar noon. Therefore, profile-based methods such as Pro-Energy are effective for the majority of days. A challenge arises when attempting to forecast solar irradiance in the presence of clouds, haze, or the change in the seasonally-elevated angle of the sun. The running correction term ρ in MPE is used to partially predict deviations from the expected solar irradiance within the same day, while the recent history of similar days is exploited by the MLP's rolling median and maximum features.

C. Why PER Is the Right Metric for This Problem

The Prediction Error Ratio is defined as:

$$\text{PER} = \frac{1}{N} \sum_{i=1}^{N} \left| 1 - \frac{\hat{E}_i}{H_i + \varepsilon} \right|, \tag{1}$$

where $\varepsilon = 10^{-5}$ is used to avoid dividing by zero. The principal characteristic of PER that makes it appropriate for energy management of WSNs is the use of $H_i + \varepsilon$ as the denominator. This means that the error is relative to the actual value for each time slot and not an absolute value. For example, an error of 0.001 W/m^2 at sunrise, when true irradiance is also around the same value, will have a significantly different operational impact than the same absolute error at solar noon when true irradiance may be 1.0 W/m^2. Therefore, an error of 0.001 W/m^2 at sunrise would correspond to a 100% relative error, resulting in the node's energy budget being completely inaccurate at that slot. However, the same absolute error during noon would only equate to 0.1%, which is insignificant. PER averages and equally weights both of these instances in terms of the effect on the battery state across the whole day. Methods optimised for absolute error (such as those trained with MSE) will typically place their emphasis on the highest irradiance time slots and will not produce an adequate level of accuracy for the boundary slots near zero irradiance. This is the same failure mode observed for CNN-LSTM-AC on Day 100 and Day 300 in this study.

IV. Proposed Methodology

A. EWMA

The Exponential Weighted Moving Average (EWMA) is the most basic baseline. It consists of only yesterday's prediction and yesterday's harvested value at the corresponding time slot, weighted by parameter α:

$$E(d, n) = \alpha\, E(d{-}1, n) + (1 - \alpha)\, H(d{-}1, n). \tag{2}$$

With $\alpha = 0.5$, it assigns equal weighting to yesterday's forecast and measurement. The recursive nature of the calculation means that anything measured two days ago will contribute at weight $(1 - \alpha)^2 = 0.25$, anything measured three days ago at $(1 - \alpha)^3 = 0.125$, and so forth, with exponentially reduced influence. The problem with this method, however, is that decay occurs regardless of whether the historical pattern of irradiance is consistent or inconsistent. Hence, if there are weather variations from recent history, EWMA will lag behind the actual signal until many averages accumulate, so it is not as responsive on anomalous days as it typically is on stable days, and will lag behind the true signal by one day in the case of a stable weather pattern.

B. WCMA

The Weather-Conditioned Moving Average (WCMA) adds new data about energy usage to the EWMA method by maintaining up to D days of information. It calculates the gap factor GAP_k by comparing the recent reading for the current day with the historical average value:

$$E(d, n) = \alpha H + (1 - \alpha)\, M_D(d, n)\, \text{GAP}_k, \tag{3}$$

where $M_D(d, n)$ is the mean energy at slot n over the past D days, and $\text{GAP}_k = V{\cdot}F \,/\, \sum F$ with $V_k = E(d, n{-}K{+}k)/M_D(d, n{-}K{+}k)$ and $F_k = k/K$. The GAP_k term is a ratio of the current day's recent readings to the average of those same readings over historical data. When $\text{GAP}_k > 1$, the current day's reading exceeds the historical mean and the forecast is adjusted upward; when $\text{GAP}_k < 1$, the current day is dimmer and the forecast is adjusted downward. The linear weights F_k give the greatest influence to the most recent observations. WCMA does produce more accurate calculations than EWMA alone; however, the gap factor must be recalculated at every slot.

C. Pro-Energy

Solar irradiance is very likely to follow similar patterns each day. To exploit this, Pro-Energy keeps track of the past D days of data and compares them to today's observations based on

MAE, then calculates a weighted average prediction from the most similar historical days:

$$\hat{E}(d,n) = \alpha H + (1-\alpha)\,\mathrm{WP}, \tag{4}$$

$$\mathrm{WP} = \frac{\sum_{j=0}^{P} W_j \cdot E_{d_j}}{P-1}, \quad W_j = 1 - \frac{\mathrm{MAE}(E_{d_j}, C)}{\sum_{j=1}^{P} \mathrm{MAE}(E_{d_j}, C)}. \tag{5}$$

Historical days with lower MAE against today's observed slots receive higher weights. The profile comparison uses only the slots that have been recorded for today at the time of creating the prediction, meaning that as the day progresses and more of today's profile is observed, the weights are updated accordingly. The principal issue is that the weighted predicted value does not take into account any errors that may accumulate throughout the day beyond this profile update, nor will it produce an accurate prediction if today's profile looks significantly different from any of the D historical profiles. An example of this is if there are unusual transient cloud cover patterns or abnormal weather phenomena that create a profile significantly different from all historical profiles; in this case, all weights would be very similar and the weighted average would revert to the historical mean, which may be far from the truth.

D. *Modified Pro-Energy (MPE)*

The introduction of MPE was done to remedy the issue of within-day error build-up in Pro-Energy [15]. It keeps track of a running correction term $\rho(d,n)$ via an exponential moving average with parameter β:

$$\rho(d,n) = \beta\,\rho(d,n{-}1) + (1-\beta)\Big[\hat{E}(d,n{-}1) - E(d,n{-}1)\Big]. \tag{6}$$

The initialisation for the correction term ρ at the beginning of each day is set to zero. At each subsequent time slot, a fraction $(1-\beta)$ of the most recent prediction error is recorded and a fraction β of the previous correction factor is carried forward. When $\beta = 0.2$, MPE is comparatively more responsive to recent errors, with approximately 80% of the new correction coming from the most recent time slot's error and only 20% from prior errors. By using this method of weighting, MPE reacts very quickly when the model begins to over-predict or under-predict during a particular day, which is ultimately what causes Pro-Energy failures.

The final MPE prediction subtracts this correction:

$$\hat{E}(d,n) = \alpha H + (1-\alpha)\,\mathrm{WP} - \rho(d,n). \tag{7}$$

When the model over-predicted in the previous time slot, the error is positive, ρ grows positive, and the next predicted value is decreased. Conversely, if the model under-predicted, the error yields a negative value, ρ grows negative, and the next predicted value is increased. The self-correcting nature of MPE allows it to progressively recover from poor initial estimates even if the weighted profile WP remains inaccurate. In our results, MPE alone achieves an average PER of 0.8976, confirming that the error-correction mechanism already solves most of the prediction problem.

E. *CNN-LSTM with Actor-Critic (CNN-LSTM-AC)*

In Nguyen et al. [26]'s research, they forecast solar irradiance for the next hour using NREL's solar irradiance data. The results are included here to show how deep learning models can be used as baselines in solar energy prediction with WSNs, to compare the performance of proposed and standard baseline models in predicted accuracy.

Base architecture: Each CNN-LSTM model uses two convolutional layers with max-pooling to extract local patterns from the input sequence, followed by an LSTM layer that captures longer dependencies across time. Ten models are trained with different hyperparameter settings. The top L models by validation loss are kept and their outputs are averaged to form an ensemble:

$$\hat{y}_{en}(t) = \frac{1}{L}\sum_{i=1}^{L} \hat{y}_i(t). \tag{8}$$

Actor-Critic refinement: The Actor-Critic (AC) structure provides a technique for fine-tuning ensemble forecast predictions through temporal difference error calculation by the Critic and a correction through scaling and translating the ensemble output by the Actor:

$$y_{pre}(t) = a_t \cdot \hat{y}_{en}(t) + b_t. \tag{9}$$

The Critic evaluates this correction using a temporal difference error and updates the Actor to minimise squared prediction error:

$$r_t = -\left(y_{pre}(t) - y(t)\right)^2. \tag{10}$$

The AC layer works as a slot-by-slot post-processing filter. The Actor outputs two scalars (a_t, b_t) from a state vector that includes the current ensemble prediction and the normalised slot index. The scale a_t is constrained to the range $(0.5, 1.5)$ so that the ensemble is never inverted, and the offset b_t is constrained to $(-0.2, 0.2)$ as a small additive correction. These constraints reflect the intended role of the AC layer as a fine-tuning step rather than a large correction.

Why CNN-LSTM-AC struggles with PER: The model was created to forecast one hour ahead of time with grid-connected systems where MAPE is utilised for evaluation purposes. PER is a much more difficult metric because it measures the relative error observed for each time interval, especially for near-zero irradiance. During the boundary minutes near sunrise and sunset, even a small absolute prediction error will contribute substantially to the total PER error due to the near-zero denominator $H_i + \varepsilon$. As a result, the CNN-LSTM-AC model, which has been optimised to minimise squared absolute error around large irradiance values, will not have the necessary pressure to improve accuracy in these boundary regions. MPE, by contrast, carries a running correction term that adapts to the actual error seen in the current day's boundary slots as they are observed, so it naturally handles those regions better.

F. *Hybrid ANN Model*

The MPE has already solved the majority of the forecasting issue with profile matching and correcting within-day errors.

The only remaining failure source is days where there is no relevant historical profile. The weighted profile WP is not a great starting point for those days, and the error correction ρ cannot recover sufficiently in the early part of the day before enough slots have been observed to build up a meaningful correction. A model based on slot-level features is less prone to this error because it does not depend on finding a good match to a previous day in the profile store.

A lightweight MLP has been used as a secondary component blended with the MPE. The MLP is trained using a 15-day rolling window so that it is calibrated to the latest irradiance regime. Because the training window is relatively small, the MLP does not train on irrelevant patterns from distant parts of the year.

Feature engineering: The feature vector consists of six features for each time slot: (i) $H(d-1, n)$ — yesterday's energy level; (ii) median energy over the past W days at slot n; (iii) maximum over the same window; (iv) $H(d-1, n-1)$ — the energy level for the previous time slot on the previous day; (v) $\sin(2\pi n/1440)$; (vi) $\cos(2\pi n/1440)$. The sinusoidal pair gives the MLP a continuous, repeating value describing what part of the day corresponds to the time slot. If just using a raw time slot index, then slot 1439 (23:59) would have a discontinuity with slot 0 (00:00); sine and cosine allow for smooth progression. The value of energy yesterday during the same time slot is the single most informative feature because it reflects the magnitude of irradiance at this last comparable point in time.

Network architecture: The MLP consists of two hidden layers with 32 and 16 neurons, uses a ReLU activation function, and L2 regularisation with $\alpha_{\text{reg}} = 0.01$. Early stopping monitors a 10% validation split with patience of 10 iterations and a maximum of 200 epochs. Inputs and outputs are scaled using Min-Max normalisation prior to training and inverse-transformed after prediction. The MLP is retrained based on the last 15 days of history for each target day. The total number of parameters in the MLP is under 600, and retraining completes in under one second on a standard CPU.

Hybrid combination: The final hybrid prediction is:

$$\hat{E}_{\text{HYB}}(d, n) = \lambda\, \hat{E}_{\text{ANN}}(d, n) + (1 - \lambda)\, \hat{E}_{\text{MP}}(d, n), \qquad (11)$$

with $\lambda = 0.2$, so MPE contributes 80% and the MLP contributes 20%. This ratio reflects how well MPE already forecasts for most days, so the MLP's 20% share will provide enough of an improvement on bad forecasting days while providing adequate stability to allow the forecasts to remain consistent for good forecasting days. When the MLP produces a poor forecasting result on a given day, the 0.2 weight will reduce its effect on the overall result. This deliberately uneven distribution means most of the accuracy comes from the MPE self-correction mechanism, and the MLP provides additional corrective input. Fig. 1 shows the overall architecture. Algorithm 1 gives the full procedure.

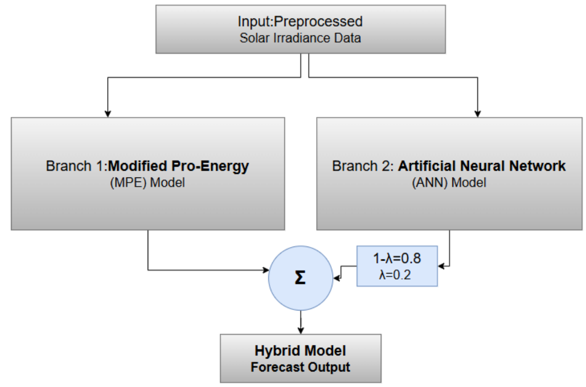

Fig. 1: Architecture of the proposed hybrid model. Solar irradiance data flows into two parallel branches: MPE and MLP. Their outputs are combined at a fixed blend ratio of λ=0.2 (20% MLP, 80% MPE).

Algorithm 1 Hybrid ANN

Input: h, day d; α=0.5, β=0.2, λ=0.2, D=10, w=7, n_{tr}=15
Output: $\hat{E}_{\text{HYB}}(d, \cdot)$

$\text{mae}[k] \leftarrow \frac{1}{N}\sum_n |h[d-k][n] - h[d][n]|, \quad k = 1 \ldots D$
$W_j \leftarrow 1 - \text{mae}[j]/\sum_k \text{mae}[k]$
$\text{WP}[n] \leftarrow \sum_j W_j\, h[d-j][n]/(D-1)$
$\rho \leftarrow 0$
for $n = 1$ **to** $N-1$ **do**
  $\hat{E}_{\text{MP}}[n] \leftarrow \alpha\, h[d][n-1] + (1-\alpha)\, \text{WP}[n] - \rho$
  $\rho \leftarrow \beta\rho + (1-\beta)\big(\hat{E}_{\text{MP}}[n-1] - h[d][n-1]\big)$
end for
$\mathcal{T} \leftarrow \{d - n_{\text{tr}}, \ldots, d-1\}$
Build $\mathbf{X}, \mathbf{y}$ from $\mathbf{f}(t, n)$ and $h[t][n]$ for $t \in \mathcal{T}$, $n \in [0, N)$, where: $\mathbf{f}(d, n) = \big[H(d-1, n),\ \tilde{H},\ H_{\max},\ H(d-1, n-1),\ \sin\frac{2\pi n}{N},\ \cos\frac{2\pi n}{N}\big]$
Fit MinMaxScaler on $\mathbf{X}$ and $\mathbf{y}$; train $\text{MLP}(32, 16)$ with early stopping
$\hat{E}_{\text{ANN}} \leftarrow \text{scaler}_y^{-1}\big(\text{MLP}(\mathbf{X}_s^{\text{test}})\big)$, clipped to $[-0.01,\ H_{\max}]$
$\hat{E}_{\text{HYB}}[n] \leftarrow \lambda\, \hat{E}_{\text{ANN}}[n] + (1-\lambda)\, \hat{E}_{\text{MP}}[n]$, $\forall n$
return $\hat{E}_{\text{HYB}}$

G. Feature Engineering Justification

The ablation study was used to validate the significance of these six features by measuring the additional prediction error of the hybrid model for the five test days with one feature removed at a time. Table I shows the PER for each configuration and Table II shows the corresponding percentage change.

TABLE I: Ablation Study: Hybrid PER When One Feature Is Removed

Configuration	D10	D50	D100	D200	D300	Avg
All features (Full)	**0.6712**	**0.6757**	0.9629	0.6959	**0.9054**	**0.7822**
w/o $H(d-1, n)$	0.6733	0.8744	0.9962	1.6128	0.9050	1.0123
w/o roll. median	0.7011	0.7842	**0.8834**	**0.6949**	0.9064	0.7940
w/o roll. max	0.6541	0.7120	0.9653	0.7047	0.9064	0.7885
w/o $H(d-1, n-1)$	0.6609	0.7090	0.8870	0.6791	0.9059	0.7684
w/o $\sin(2\pi n/N)$	0.6618	0.8295	0.8881	0.7697	0.9057	0.8110
w/o $\cos(2\pi n/N)$	0.6784	0.8155	0.9154	0.6806	0.9054	0.7991

The most significant feature, by a large margin, is yesterday's energy $H(d-1, n)$, which increased the average PER by 32.98% when removed. The largest single-day change in PER was observed on Day 200, where the removal of this single

TABLE II: Percentage PER Change When Each Feature Is Removed (positive values mean the feature was helping)

Feature removed	D10	D50	D100	D200	D300	Avg Δ%
$H(d-1, n)$	$+0.31$	$+29.41$	$+3.46$	$+131.76$	-0.04	$+32.98$
Roll. median	$+4.45$	$+16.06$	-8.26	-0.14	$+0.11$	$+2.44$
Roll. max	-2.55	$+5.37$	$+0.25$	$+1.26$	$+0.11$	$+0.89$
$H(d-1, n-1)$	-1.53	$+4.93$	-7.88	-2.41	$+0.06$	-1.37
$\sin(2\pi n/N)$	-1.40	$+22.76$	-7.77	$+10.60$	$+0.03$	$+4.85$
$\cos(2\pi n/N)$	$+1.07$	$+20.69$	-4.93	-2.20	0.00	$+2.93$

feature caused the PER to increase from 0.6959 to 1.6128 — an increase of 131.76%. On Day 200, the weather was likely transitional and no historical profiles matched the forecast well, making the most recent same-slot reading the most valuable signal available. The cyclical features $\sin(2\pi n/N)$ and $\cos(2\pi n/N)$ generated an average increase of 4.85% and 2.93% respectively when removed. The greatest impact of removing the sine feature was on Day 50 and Day 200, with increases in PER of 22.76% and 10.60% respectively. The irradiance on both days had a strong morning rise and a steady afternoon decline, and thus the MLP required a compact time signal to model the data accurately.

Over the days, the rolling median provides a very inconsistent effect; it is beneficial on Day 10 and Day 50, while Day 100 and Day 200 experience a subtle disadvantage. This could be due to the rolling median generating an inaccurate estimate for Day 100, since all prior days were typically normal whereas Day 100 was not, thereby pulling the MLP forecast toward the historical norm rather than the current anomalous conditions. On all of the normal days, it serves as a stable reference point that helps reduce noise between individual slots. The rolling maximum has a smaller and more consistent effect, contributing positively at less than 1% on average. All six features are retained in the final model; none add any substantial computational cost, and no single feature consistently hurts performance across all five test days.

V. Dataset and Preprocessing

The Solar Radiation Research Laboratory (SRRL) at the National Renewable Energy Laboratory (NREL) [15] is the source for the solar irradiance dataset. This dataset consists of data from 12 different sensors at 1-minute intervals in units of W/m^2. This is the same dataset that is utilised in evaluating the original Pro-Energy model as demonstrated by Cammarano et al. [15]. The ability to make direct comparisons between the two models by using the same data allows the researchers to make valid and meaningful comparisons.

The Global CMP22 channel is used as the prediction target. This channel provides an estimate of global solar radiation using a broadband measurement. The reason the CMP22 was used in the original Pro-Energy work was that the CMP22 pyranometer measures total incoming solar radiation across the entire spectrum; therefore, the energy available to the photovoltaic cell corresponds most closely to that supplied by the CMP22. If a directional or spectrally narrow measurement were used, additional conversion factors would need to be applied, limiting generalisation across other panel types.

The dataset consists of 366 days, each with 1440 one-minute time slots, providing a total of 527,040 data points. The high-resolution 1-minute data is critical for evaluating the PER metric. If boundary-region errors were averaged over the full first or last hour of the day, they would be diluted; in contrast, when 1-minute data is used, each boundary slot contributes equally to the overall PER average. Thus, the final PER values from each model are directly comparable to those reported in the original Pro-Energy paper, which used the same resolution and dataset.

Preprocessing steps: (1) read the CSV file and rename the columns; (2) convert all dates into integer day indices; (3) convert time strings to minute integer values; (4) construct a two-dimensional matrix h of shape 366×1440 where $h[d][n]$ holds the irradiance at minute n on day d; (5) apply Min-Max normalisation to inputs and outputs during ANN training only, with the scaler fitted on the training window and applied to the test features. No normalisation is applied to the statistical methods, which operate directly in W/m^2. Table III shows a short sample from the dataset.

TABLE III: Sample Rows from the NREL Dataset (Day 0)

Date	MST (min)	CMP22 (W/m^2)
0	436	8.428
0	437	9.574
0	438	10.391
0	439	10.082
0	440	10.131

These rows correspond to the early-morning ramp at about 07:16 MST on the first day of the dataset. CMP22 measurements went from 8.428 to 10.131 W/m^2 in five minutes, demonstrating the gradual change typical of clear days during sunrise. Within this region PER is most affected because of the small denominator in the PER equation, so that even small absolute prediction errors can generate significant relative results. The maximum recorded irradiance for the dataset was 1.248 W/m^2, recorded near solar noon on a clear summer day.

VI. Experimental Setup

All the code was developed using Python 3.11. The MLP and statistical methods used NumPy, Pandas, Matplotlib, and scikit-learn, while the CNN-LSTM-AC model used PyTorch with CUDA acceleration running on an NVIDIA RTX 3060 GPU. All six methods were run on the same data and evaluated with the same PER formula. The hyperparameters for all methods can be found in Table IV.

The five test days selected for evaluation were Day 10, Day 50, Day 100, Day 200, and Day 300. Each of these days represented different seasons and weather throughout the year. Day 10 was in January, relatively early in the year with low sun angles and shorter daylight hours. Day 50 was at the end of February or beginning of March, representing early spring conditions. Day 100 was in April, a month when the weather is usually variable at the NREL Colorado location. Day 200 was in mid-July, the height of summer with the maximum sun angle and longest day length. Day 300 was

TABLE IV: Hyperparameter Settings for All Methods

Parameter	Value
EWMA: α	0.5
WCMA: α, D, K	0.5, 5 days, 5 slots
Pro-Energy: α, profiles	0.5, 10 days
MPE: α, β	0.5, 0.2
ANN: hidden layers	(32, 16), ReLU
ANN: L2 regularisation	0.01
ANN: training window	15 days
ANN: feature window w	7 days
ANN: early stopping patience	10 iterations
ANN: max epochs	200
Hybrid blend λ	0.2
CNN-LSTM-AC: Conv layers	2 (32 and 64 filters)
CNN-LSTM-AC: LSTM units	64
CNN-LSTM-AC: ensemble models L	top 10 by val. loss
CNN-LSTM-AC: epochs	300
CNN-LSTM-AC: batch size	16
CNN-LSTM-AC: dropout	0.1 to 0.6

at the end of October and in the transition to autumn with the lowest sun angles and shortest day length. This spread ensures that methods are tested across a full range of irradiance magnitudes, profile shapes, and seasonal transition effects.

The evaluation metric is:

$$\text{PER} = \frac{1}{N}\sum_{i=1}^{N}\left|1 - \frac{\hat{E}_i}{H_i + \varepsilon}\right|, \tag{12}$$

where H_i is the actual irradiance at slot i, $\hat{E}_i$ is the prediction, and $\varepsilon = 10^{-5}$ prevents division by zero when $H_i \approx 0$. PER is the metric used by Cammarano et al. for Pro-Energy [15] and is appropriate for a WSN because it penalises relative errors at every minute of the day, including low-light boundary periods.

VII. Results and Analysis

A. Prediction Curves

All six prediction curves on Day 10 (slots 500–900) are shown in Fig. 2. The ANN Hybrid remained close to the true ground truth throughout this time frame. EWMA experienced more variability in its predicted values than actually occurred and over-predicted on a consistent basis. While WCMA was more stable than EWMA, it lagged behind during periods of rapid irradiance change. Pro-Energy and MPE made reasonable predictions during periods of lower variability; however, they do not perform as well during periods of high variability. Although the CNN-LSTM-AC model tracked the overall shape well, there was significantly greater variability in predicted values between slots than the hybrid model.

Fig. 3 displays a zoomed-in view of the declining portion of Day 10 (slots 600–700). This transition period serves as an opportunity to compare how well and quickly these methods adjust to dropping irradiance values. Within this transition timeframe, the ANN Hybrid was able to follow the declining signal better than both the CNN-LSTM-AC and MPE models. The MLP piece of the ANN Hybrid model had learned from the rolling features just how quickly a day usually ends, and that knowledge is reflected in its final prediction value.

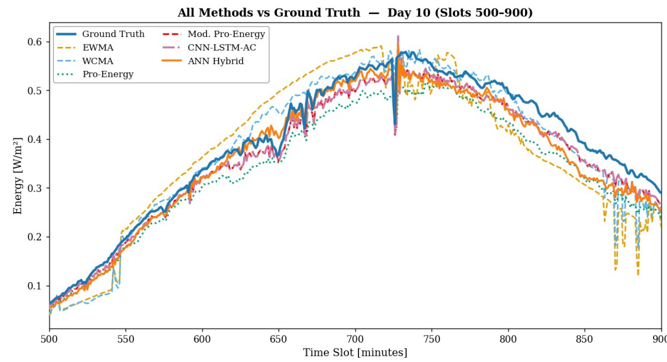

Fig. 2: Prediction curves for Day 10 (slots 500–900). All six methods are shown against the ground truth. The ANN Hybrid tracks the actual measurements most closely throughout.

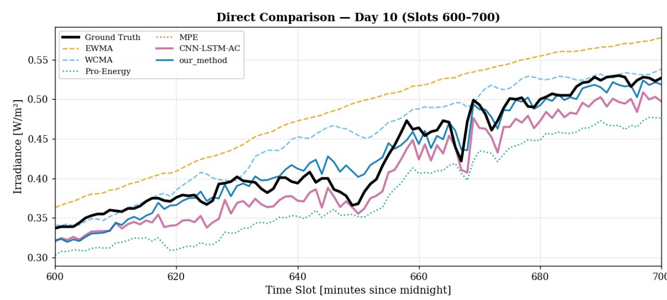

Fig. 3: Declining phase of Day 10 (slots 600–700). The ANN Hybrid follows the ground truth more closely than CNN-LSTM-AC and MPE.

B. PER Results

Table V shows PER for all six methods on all five test days.

TABLE V: PER Across All Methods and Test Days (lower is better)

Day	EWMA	WCMA	Pro	MPE	CNN-AC	Hybrid
Day 10	1.2546	1.0939	0.9338	0.8846	0.8846	**0.6712**
Day 50	1.4395	1.0067	5.9780	0.8118	0.6447	**0.6757**
Day 100	1.7872	3.2683	1189.86	1.1594	365.6113	**0.9629**
Day 200	1.1061	0.9839	0.9759	0.7141	0.6852	**0.6959**
Day 300	3.0204	1.3640	0.9315	0.9179	121.9912	**0.9054**
Avg	1.7216	1.5434	239.74	0.8976	97.9634	**0.7822**

The hybrid achieves the lowest PER on every test day and the best average of 0.7822. Several points from the table are worth examining in detail.

Day 10: Each of the methods produced acceptable, finite PER values for Day 10 since it has the characteristics of being a winter day with relatively low variability. The ranking of the methods for this day is Hybrid $<$ CNN-AC $=$ MPE $<$ Pro $<$ WCMA $<$ EWMA. This is expected since it is a day of little variability: the hybrid benefits from MLP-derived day-to-day similarity, and the CNN-AC performs on the same level as MPE since January historical days provide almost exact matches, and the lack of multi-day context is the reason for EWMA's relatively poor performance. The difference in PER between EWMA (1.2546) and the hybrid (0.6712) is 46.5%, which is the hybrid's smallest advantage over EWMA across all test days, indicating that the short memory of EWMA is least detrimental during stable weather days.

Day 50: Pro-Energy produced a PER of 5.9780 for Day 50, which is greatly exceeded by all of the other methods. The result for Day 50 indicates that the irradiance profile is considerably different from any of the ten historical profile shapes for Days 40–49. The weights of all profiles are approximately the same, meaning a weighted average WP returns back to the mean shape of the profiles even though this may differ from the true shape on Day 50. MPE recovers from this by using an error correction loop to arrive at a final result of 0.8118. CNN-LSTM-AC achieves its best result of 0.6447 on Day 50 because it has seen the profile of early spring during its training phase. The hybrid reaches 0.6757, staying close to CNN-LSTM-AC with no GPU required.

Day 100: This is the best case for illustrating the performance level of each model. Pro-Energy produces a PER of 1189.86 because Day 100's irradiance profile does not resemble any of the historical data. Because the weighted average WP was incorrect from the beginning of the day, Pro-Energy has no way to recover from this error. CNN-LSTM-AC also fails badly with a PER of 365.61, due to use of historical sliding window training days which did not contain any representative day for this anomalous pattern. The Actor-Critic layer can only make linear adjustments to the ensemble mean and is unable to perform any type of recovery when the ensemble's average is far from the truth. WCMA also struggles, reaching 3.2683, as the GAP factor magnifies the current mismatch. MPE's running correction term ρ is effective in utilising cumulative error to correct the forecasted values, effectively pushing the forecast closer to the actual values, reaching 1.1594. The hybrid reaches 0.9629, with the MLP's slot-level features, particularly yesterday's same-slot energy, providing an independent basis for a more reasonable estimate.

Day 200: Being the only summer day in the test set, this day demonstrated the fewest variances among all datasets. Each algorithm with multi-day context performed adequately. CNN-LSTM-AC produced its second-best result at 0.6852. MPE reached 0.7141 and the hybrid 0.6959. Interestingly, the ablation study showed that removing $H(d-1, n)$ on Day 200 raised the hybrid's PER to 1.6128 — the largest effect of any single feature in the study. This confirms that there existed significant day-to-day variability in magnitude on this day, and that the highest correlation with the present forecast was due to the previous day's same-slot reading.

Day 300: CNN-LSTM-AC has a performance similar to Day 100, reaching 121.99. Day 300 occurs later in the year when many near-zero irradiance slots exist due to low sun angles. The presence of large numbers of near-zero irradiance slots results in a dramatic increase in PER when predictions are in error. CNN-LSTM-AC had no slot-by-slot correction mechanism for observed errors within the day, so the result is a very large PER. Pro-Energy performs well here at 0.9315, slightly better than MPE (0.9179), because both had access to the same low-irradiance profiles from Days 290–299 and the profile matching worked correctly for this predictable autumn pattern. The hybrid system's performance is slightly better at 0.9054, due to the support provided by the MLP.

Why MPE dominates the PER competition: PER measures relative error at all slots, including those at the boundary of near-zero irradiance. Therefore, when H_i is near zero, even a very small absolute prediction error will produce a very large PER contribution. The ρ correction in MPE accumulates every observation of error minute by minute and applies a corresponding opposite correction to the next prediction. This is precisely what is required for the boundary slots: should the model have over-predicted in the previous minute, ρ will be positive and reduce the subsequent prediction, moving it towards zero. CNN-LSTM-AC does not have an equivalent mechanism. The Actor-Critic layer works at the level of the ensemble output and therefore cannot make minute-by-minute corrections based on what has already been observed today.

C. Average PER and Percentage Improvement

The average PER achieved by the six methods is shown in Fig. 4. The hybrid method had a clear advantage, producing a significantly lower average PER than the other methods. The Pro-Energy and CNN-LSTM-AC methods performed poorly on Days 100 and 300.

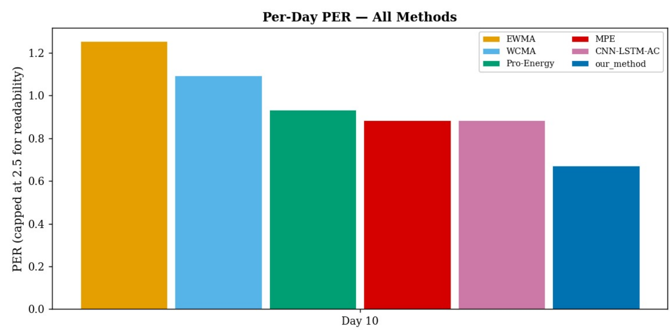

Fig. 4: Average PER across all five test days for each method. The ANN Hybrid produces the lowest average PER. Note that Pro-Energy and CNN-LSTM-AC averages are dominated by their failures on Day 100 and Day 300.

Fig. 5 shows how much better the ANN Hybrid performed than the baseline methods for each day of testing. The improvement of the ANN Hybrid over each of the baseline methods was 46.5% (EWMA), 38.6% (WCMA), 28.1% (Pro-Energy), and 24.1% (MPE) on Day 10. The improvements over Pro-Energy and CNN-LSTM-AC on Day 100 were above 99% due to the complete failure of those methods. On Day 200, the ANN Hybrid showed a 28.7% improvement over Pro-Energy and a 2.5% improvement over MPE. On Day 300, Pro-Energy and MPE produced similar levels of improvement (2.8% and 1.4% respectively), whereas CNN-LSTM-AC again failed completely, with the ANN Hybrid showing an improvement of over 99% over it.

D. The Role of MPE and the Role of the MLP

The MPE's performance (average PER 0.8976) clearly demonstrates that MPE already outperforms every other method except the hybrid. This shows that MPE's error correction term provides an adequate correction for errors that would have been a challenge for purely learning-based models

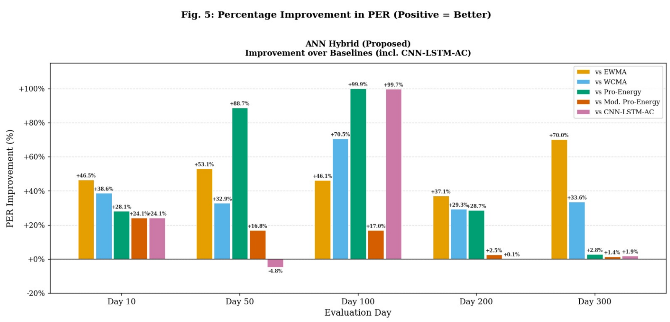

Fig. 5: Percentage improvement of the ANN Hybrid over each baseline method per test day. Positive values mean the Hybrid has lower PER. The largest gains are on Day 100 and Day 300 where Pro-Energy and CNN-LSTM-AC fail completely.

to achieve by themselves. The MLP's job is not to supplant MPE, but to provide a better initial estimate for the current day when all historical profiles had poor results.

For example, on Day 100, MPE started with a poor profile estimate and had to spend the majority of early-day hours correcting it through ρ. The ρ correction continues to build over the course of the next several hundred slots, meaning early-morning predictions may have been significantly in error until there was enough volume in ρ to compensate. The MLP, using the previous day's same-slot readings and rolling statistics, supplied a better starting estimate. Therefore, the MLP substantially reduced the number of in-day corrections needed on Day 100. The 80/20 approach allows the MLP to have just enough impact to slightly shift the starting point without completely taking over from MPE's self-correcting loop.

On days like Day 200 and Day 300, MPE has already done most of the work using its profile matching; therefore, throughout the entire day, ρ remains small. With this situation, the MLP is only going to provide 20% of input, and therefore the small changes from the MLP over the baseline MPE are reflected in the 2.5% and 1.4% improvements respectively. The overall construction of the hybrid is a relatively conservative approach: the MLP provides assistance only when MPE requires it, and avoids interfering when MPE does not. Table VI summarises the main differences between the hybrid and CNN-LSTM-AC.

TABLE VI: Comparison: ANN Hybrid vs. CNN-LSTM-AC

Aspect	ANN Hybrid (Ours)	CNN-LSTM-AC
Architecture	Shallow MLP + MPE	Deep CNN + LSTM + RL
Total parameters	<600	~200K per model
Training data needed	15-day rolling window	40 to 55 day sequences
GPU required	No	Yes
Metric optimised for	PER (WSN-specific)	MSE / MAPE (grid)
Within-day correction	Yes (via ρ)	No
Avg PER	**0.7822**	97.9634
Stable on hard days	Yes	No

VIII. Discussion

A. The Importance of Within-Day Correction for PER

Correcting errors within a day has a more pronounced impact on the overall PER value than the complexity of the model used to calculate that value. However, this conclusion must be viewed carefully as it relates only to PER's characteristics in relation to energy management in WSNs. The formula for PER is based on weighting the prediction errors at each time slot relative to the actual irradiance for that slot. This results in the total PER for a complete day being affected more by the boundary time slots close to sunrise and sunset, when actual irradiance is close to zero. For example, an error of 0.001 W/m^2 at sunrise when the true value is 0.0001 W/m^2 contributes a PER term of $|1 - 0.001/(0.0001 + 10^{-5})| \approx 9.1$, whereas the same absolute error at solar noon (when true irradiance is around 1.0 W/m^2) contributes only $|1 - 1.001/1.0| \approx 0.001$ to the sum. For a method to have a low PER value, it must correctly predict the irradiance for the boundary time slots.

By using the running correction ρ to carry the memory of recent prediction errors forward in time, MPE will have had several hundred time slots to adjust the correction value as the day progresses towards sunrise. If the model has been over-predicting at dawn, ρ will grow strongly positive and push the prediction towards zero well before it hits any individual boundary slot. On the other hand, CNN-LSTM-AC's Actor-Critic layer adjusts (a_t, b_t) based on the squared error at the previous slot, but on a day like Day 100 the entire ensemble output is completely inaccurate from the first slot onward, and the linear correction cannot compensate for a fundamentally wrong ensemble shape.

This analysis has implications for future work: any prediction method to be evaluated under PER for WSN applications should include some type of within-day correction mechanism, whether it is an exponential correction (e.g., MPE), a Kalman filter update, or a slot-by-slot residual correction. The lack of such correction methods will lead to even a sophisticated model being unable to compete against simpler methods that include within-day correction.

B. Method Selection Guidance

When a practitioner decides on a prediction method for a new WSN deployment, the following considerations apply based on the results:

- An MPE method is likely to be adequate if the deployment site has a stable and predictable climate with no anomalous days.
- If the deployment site anticipates seasonal transitions or irregular days with unusual patterns, the hybrid method provides additional robustness at the cost of a rolling 15-day retraining step that takes under one second on a CPU or gateway node.
- CNN-LSTM-AC is not advised for deployments where PER is the primary evaluation criterion or where GPU hardware is unavailable.
- Pro-Energy is not appropriate for deployment sites where an anomalous day would create an unacceptable operational impact because it has unbounded failure modes

— a PER greater than 1000 means the predicted value is orders of magnitude higher than the actual irradiance, which could result in a battery becoming depleted in a matter of hours.

- EWMA and WCMA should only be considered as sanity-check baselines. Their average PER values of 1.72 and 1.54 imply that they systematically deviate by more than 100% from the actual irradiance on average across all time slots, making them unsuitable for tight energy-neutral scheduling.

IX. Conclusion

The study introduced a hybrid solar power forecasting approach that blends Modified Pro-Energy and a lightweight MLP neural network, evaluated on the NREL solar irradiance dataset using PER as the comparison standard [15]. In total, the research compared five different forecasting methods including EWMA, WCMA, Pro-Energy, MPE, and CNN-LSTM-AC from Nguyen et al. [26].

The hybrid achieved the lowest PER on all five test days and the best average PER of 0.7822. MPE alone reached 0.8976, confirming that the within-day error correction is the most valuable part of the design. The MLP's contribution was to furnish improved starting estimates on days when no historical profiles matched well, allowing the hybrid to reduce the overall effort of correction — improving over MPE by approximately 16.9% on Day 100 and staying within 1.4% of MPE on Day 300.

CNN-LSTM-AC, despite its much more complex architecture with around 200K parameters per model and GPU requirements, produced an average PER of 97.9634, driven by complete failures on Day 100 (PER 365.61) and Day 300 (PER 121.99). The core reason is that CNN-LSTM-AC has no within-day correction mechanism and is optimised for squared absolute error rather than PER, and the model is unable to generalise when the current day's pattern does not appear in any training window. The ablation study revealed yesterday's same-slot energy to be the most significant input feature, providing a 32.98% average PER improvement when included. The cyclic time features $\sin(2\pi n/N)$ and $\cos(2\pi n/N)$ were the next most significant, providing average improvements of 4.85% and 2.93% respectively.

The hybrid model has under 600 parameters, retrains in under one second on a CPU, and requires no GPU. For WSN nodes where hardware limitations are at their most stringent, training can be done at a gateway and only inference weights sent to the node. The evaluation covers five test days at a single geographic location, and the fixed blend weight $\lambda = 0.2$ and hyperparameters were not tuned on a held-out validation set, so generalisation to other climates and deployment sites remains to be confirmed. Future work will test the model on geographically diverse datasets, investigate adaptive blend weights that respond to recent prediction confidence, and integrate the predictor with an energy-neutral scheduling policy so that prediction uncertainty feeds directly into duty-cycle decisions.

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